TEXORA · LECTURE SHEET

Fabric Cutting

Severing the lay along the marker's lines so that every ply yields identical, accurate panels: the requirements of good cutting, the process step by step, the machines from hand shears to the CNC cutter, quality control of the cut work, and the sorting and bundling that hand the panels to sewing.

TE-2207 · APPAREL MANUFACTURING-1 · TOPIC 6 · LEVEL 2 TERM 2

1. What cutting is

DEFINITION

Cutting is the separation of the spread lay into garment panels by cutting along the outlines of the marker, so that each ply produces the complete set of pieces the marker contains. It is the first irreversible operation of garment production: every earlier error can still be corrected, but a wrongly cut lay is waste fabric.

Because one pass of the knife cuts every ply at once, cutting multiplies both correctness and error exactly as spreading does. The requirements below are the standard the cut is judged against, at inspection of the first and last plies.

2. Requirements of cutting

- **Accuracy to the line.** Cut exactly on the marker outline: cutting outside leaves panels oversize and seams bulky; cutting inside loses measurement that no sewing can restore. Spec tolerances (Topic 2) are consumed here first.
- **Identical plies, top to bottom.** The blade must stay vertical and the lay must not shift, so ply 1 and ply 120 are the same size. A leaning blade or a dragged lay makes the bottom plies systematically different.
- **Clean, unfused edges.** A sharp, correctly chosen blade leaves edges that do not fray; on synthetic fabrics the blade's friction heat can melt plies together (fused edges) if the lay is too high or the blade dull — fused panels tear apart damaged.
- **Correct notches and drills.** Notches cut to the right depth (too deep weakens the seam line, too shallow disappears) and drill marks placed exactly — the sewing room assembles by these marks.

- **Checks and stripes preserved.** Matched patterns must flow across panels as the marker planned; cutting drift destroys matching invisibly until the garment is sewn.
- **Safety.** Powered blades and a tall, unstable lay are the cutting room's main hazard: guards, mesh gloves for hand-knife work, clamps before cutting, and no hands ahead of the blade — ever.

3. The cutting process, step by step

1. **Check before cutting:** marker against the cut plan (style, sizes, quantities), marker placed square on the lay, grain direction confirmed, lay ends and edges aligned.
2. **Secure the lay:** clamps at the ends (or vacuum on a CNC table) so the stack cannot shift under the knife.
3. **Rough cut (blocking out):** straight separations divide the lay into convenient blocks around groups of pieces, making the work movable and the precision cuts shorter.
4. **Precision cut:** each piece is cut to its exact outline, curves (neck, armhole, crotch) taken slowly with the blade guided on the line; small pieces are often finished on the band knife for accuracy.
5. **Notch and drill** at the marked positions.
6. **Remove and check:** waste is stripped away; the top and bottom plies of a main piece are compared against the pattern — the standard first check of any cutting.

4. Methods and machines of cutting

MACHINE / METHOD	PRINCIPLE	BEST USED FOR
Hand shears	Scissor action by hand, one or few plies.	Samples, single garments, tiny repair cuttings; no production lays.
Straight knife	A vertical blade reciprocating up and down, driven by a motor on a stand that slides under the lay; the machine travels through the fabric (Figure 1).	The general production cutter: medium to high lays, most fabrics, straight and moderately curved lines.
Round (rotary) knife	A circular blade rotating against a base plate, rolling through the fabric.	Low lays and single plies, long straight cuts and gentle curves; sampling rooms.
Band knife	A continuous loop blade running over wheels,	Precision cutting of small pieces and sharp curves from pre-blocked sections; accuracy higher than hand-guided

	stationary; the operator moves the fabric block against the blade.	machines.
Die cutter (clicking press)	A shaped steel die pressed through the lay by a hydraulic press, stamping the piece in one stroke.	Large quantities of identical small parts — collars, cuffs, pockets, interlinings.
CNC (computerised) cutter	The digital marker drives a reciprocating knife head over a vacuum table that holds the lay flat and still.	High-accuracy production fully integrated with CAD (Topic 7); single-ply to multi-ply.
Laser / water-jet / ultrasonic	A beam or jet severs the material without a mechanical blade (laser seals synthetic edges as it cuts).	Single-ply and technical work, intricate shapes; niche in bulk apparel because of speed and cost.

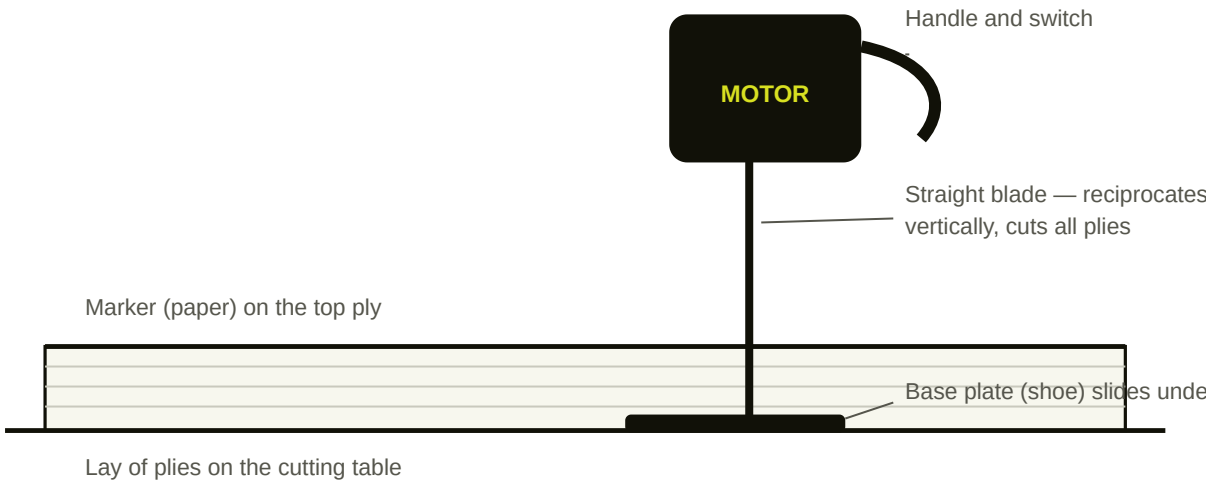


Figure 1 — The straight-knife cutting machine at work on a lay (side view, schematic).

The comparison that decides machine choice is three-way: **lay height** (straight knife highest, round knife lowest), **accuracy** (band knife and CNC above hand-guided machines, since the work or the head moves under control rather than by eye alone), and **piece size** (dies win when the same small piece repeats in thousands). Most factories run a combination: straight knife for blocking and main pieces, band knife for precision smalls, dies for interlinings.

5. Quality in cutting

CUTTING FAULT	CAUSE AND CONSEQUENCE
Frayed, scalloped edges	Dull or wrong blade for the fabric; edges unravel in sewing and seams weaken.
Fused edges	Blade heat melts synthetic plies together (lay too high, dull blade, slow travel); panels tear when separated.
Top-to-bottom size drift	Blade leaning or lay shifting mid-cut; bottom-ply garments fail measurement.
Over-cut / under-cut	Blade off the marker line; measurement loss or bulk at seams.
Notch errors	Missing, misplaced or over-deep notches; sewing assembles wrongly or seam is weakened.
Broken matching	Checks/stripes drift during cutting; panels cannot be matched at sewing.

Cutting quality is controlled at three points: before (marker and lay check, §3 step 1), during (blade sharpness, lay security), and after — by measuring panels: the top, middle and bottom ply of a principal piece are measured against the pattern/spec, and any splice-containing panels noted during spreading (Topic 5) are inspected individually. Defective panels are replaced by re-cutting from remnant fabric *of the same shade lot* before the bundle is closed.

6. Sorting and bundling — handing over to sewing

Cutting does not end at the knife. The panels must reach sewing organised so that the right pieces meet, in the right shade, in workable quantities:

1. **Numbering (ticketing).** Each panel is marked with its lay and ply number — by sticker, stamp or chalk. Pieces sewn into one garment must come from the *same ply*, because shade varies subtly from ply to ply; numbering is the only reliable way to guarantee it.
2. **Sorting.** Panels are sorted by component, size and shade: all fronts together, all backs together, bundles kept shade-pure.
3. **Panel inspection and replacement.** Damaged, fused or defective panels are replaced from same-lot remnants before bundling (§5).
4. **Bundling.** The components for a set of garments are tied into bundles of a planned quantity (commonly 10–30 garments' worth, by order size and operation). Each bundle carries a **bundle ticket**: style, size, colour, bundle number, quantity and the operations it will pass through. Bundles are the production unit of the sewing floor — output, quality and wages are all counted in bundles.
5. **Storage and issue.** Bundles are racked by style/size and issued to sewing lines in sequence; cut work waiting too long (especially white and pastel fabrics) is covered against dust and light.

7. Glossary

TERM	MEANING
Blocking out	The rough straight cuts dividing a lay into movable blocks before precision cutting.
Fused edge	Plies melted together at the cut edge by blade heat on synthetic fabric.
Bundle	A tied set of panels for a planned quantity of garments, travelling through sewing with its ticket.
Ticketing	Marking each panel with lay/ply identification so same-ply components are sewn together.
Die	A shaped steel blade that stamps out an identical piece under a press.

8. Key points and summary

- Cutting converts the lay into panels in one irreversible pass; its requirements are line accuracy, identical plies top to bottom, clean unfused edges, correct notches/drills and preserved matching.
- Process: check and secure → block out → precision cut → notch/drill → strip waste → compare top and bottom plies with the pattern.
- Machines divide by lay height, accuracy and piece size: straight knife for general production, band knife for precision smalls, dies for repeated small parts, CNC for integrated accuracy.
- Cut quality is inspected by measuring panels from top, middle and bottom of the lay; defective panels are replaced from the same shade lot.
- Numbering, sorting and bundling hand sewing a controlled flow: same-ply components together, shade-pure bundles, every bundle ticketed.

9. Practice

Answer in your own words; brief model answers follow.

1. A lay of 120 plies is to be issued in bundles of 20 garments. One cutting of one size is made from the whole lay. How many bundles of that size result, and what five items of information must the bundle ticket carry?
2. The bottom-ply shirts of a cutting are consistently 0.6 cm narrower than the top-ply shirts. Name the likely cause and the point in the process where it should have been prevented.
3. A factory cuts 5,000 identical collar interlinings a day and complains of slow, variable cutting. Which machine do you recommend, and why does it beat the straight knife here?

4. Synthetic-fabric panels are found fused at the edges after cutting. Give three contributory causes and one remedy for each.
5. Why is “replace the damaged panel from any similar fabric in the store” a wrong instruction, and what is the correct source of replacements?
6. Explain why numbering panels by ply is a shade-control measure, not merely an organisational one.

Answer key

1. 120 ÷ 20 = 6 bundles. Ticket: style, size, colour (shade), bundle number, quantity — plus the operation route where used.
2. The blade was leaning (or the lay dragged as the machine travelled), so lower plies were cut inside the line. Prevention is at the cutting stage itself: vertical blade discipline, secured/clamped lay, sharp blade needing less force — and detection is the top-vs-bottom ply comparison before bundling.
3. A die cutter (clicking press): the piece is small, identical and repeated in thousands, so one press stroke stamps it exactly every time — faster and more consistent than guiding a knife around thousands of small outlines.
4. Causes and remedies: lay too high for the blade (reduce ply height); dull blade generating friction heat (sharpen/change blade, use a waved or low-friction blade); travel too slow so heat accumulates (steady correct speed; allow blade cooling between blocks).
5. Because a panel from another fabric lot will differ in shade and possibly handle, and the mismatch appears only in the finished garment. Replacements must be re-cut from remnant fabric of the same fabric lot/shade as the lay.
6. Shade varies finely from ply to ply within a lay. If a front from ply 12 is sewn to a back from ply 87, the two halves of one garment can differ visibly. Ply numbering keeps all components of a garment from the same ply, so any ply-to-ply variation never appears inside one garment.

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